Theorem:

if $A(t)$ is UES, then for any bounded, unif PD $Q(t)$, there exists a bounded, unif. PD $P(t)$ such that

$$\dot{P}(t) + A^T(t)P(t) + P(t)A(t) = -Q(t)$$

Pf: By construction, define

$$P(t) = \int_t^\infty \Phi^T(\tau, t) Q(\tau) \Phi(\tau, t)\, d\tau$$

a.) $P(t)$ is uniformly pos-def:

$$x^T P(t) x = \int_t^\infty (\Phi x)^T Q(\tau)(\Phi x)\, d\tau$$

Define $z = \Phi(\tau, t)x$

Note since $\Phi(\tau, t)$ is nonsingular for any t, τ, $z = \Phi(\tau, t)x \neq 0$ for any $x \neq 0$

Then, for any nonzero x

$$x^T P(t) x = \int_t^\infty z^T Q(\tau) z\, d\tau$$

$$\geq \int_t^\infty \rho \|z\|_2^2\, d\tau$$

$$> 0$$

since $\|z\| > 0$, and

$$\rho = \min_t \lambda_{min}(Q(t)) > 0$$

b.) $P(t), Q(t)$ satisfy Lyap Eqn

Note that

$$\dot{P}(t) = \frac{\partial}{\partial t} \int_t^\infty \Phi^T(\tau, t) Q(\tau) \Phi(\tau, t)\, d\tau$$

$$\dot{P}(t) = \frac{\partial}{\partial t}\int_t^\infty \Phi^T(\tau,t)Q(\tau)\Phi(\tau,t)\,d\tau$$

Using Leibniz' rule:

$$\dot{P} = -\left[\Phi^T(\tau,t)Q(\tau)\Phi(\tau,t)\right]\Big|_{\tau=t} + \int_t^\infty \frac{\partial}{\partial t}\left[\Phi(\tau,t)^T\right]Q(\tau)\Phi(\tau,t)\,d\tau$$

$$+ \int_t^\infty \Phi^T(\tau,t)Q(\tau)\left[\frac{\partial}{\partial t}\Phi(\tau,t)\right]d\tau$$

recall $\frac{\partial}{\partial t}\Phi(\tau,t) = -\Phi(\tau,t)A(t)$ (Diff'n wrt 2nd time arg to STM)

and $\Phi(t,t) = I$ for any t

Then

$$\dot{P} = -IQ(t)I - \int_t^\infty A^T(t)\Phi^T(\tau,t)Q(\tau)\Phi(\tau,t)\,d\tau - \int_t^\infty \Phi^T(\tau,t)Q(\tau)\Phi(\tau,t)A(t)\,d\tau$$

$P(t)$ $\qquad$ $P(t)$

$$\Rightarrow \dot{P}(t) = -Q(t) - A(t)^T P(t) - P(t)A(t)$$

or

$$\dot{P}(t) + A^T(t)P(t) + P(t)A(t) = -Q(t)$$

c.) $P(t)$ is bounded for any t:

$$P(t) = \int_t^\infty \Phi^T(\tau,t) Q(\tau) \Phi(\tau,t)\, d\tau$$

$$\Rightarrow \|P(t)\|_i \le \int_t^\infty \|\Phi(\tau,t)\|_i^2 \|Q(\tau)\|_i\, d\tau$$

$Q(t)$ Bdd $\Rightarrow \|Q(\tau)\|_i \le \gamma_Q < \infty \quad \forall \tau$

So

$$\|P(t)\|_i \le \gamma_Q \int_t^\infty \|\Phi(\tau,t)\|_i^2\, d\tau \le \gamma_Q c^2 e^{2\lambda t} \int_t^\infty e^{-2\lambda\tau}\, d\tau$$

If $A(t)$ is UES, then

$\|\Phi(\tau,t)\| \le c e^{-\lambda(\tau-t)}$ for some $\lambda > 0$, and some $c < \infty$

So

$$\|P(t)\|_i \le \frac{-\gamma_Q c^2}{2\lambda} e^{2\lambda t} \left[e^{-2\lambda\tau} \right]_{\tau=t}^{\tau=\infty} = \frac{-\gamma_Q c^2}{2\lambda} e^{2\lambda t} \left[0 - e^{-2\lambda t} \right]$$

Thus $\|P(t)\|_i \le \frac{\gamma_Q c^2}{2\lambda} < \infty \quad \forall t.$

Thus for UES $A(t)$ (and, with a slight modification of above argument, for isL $A(t)$) we are guaranteed the existence of $P(t), Q(t)$ pairs that will work to show stability.

In fact, if there exists one $P(t), Q(t)$ pair that works there exist infinite many other pairs that will work

For LTI systems, we can take $P(t) = \text{const}$, and $\dot{P}(t) = 0$, so the Lyap Eq'n becomes

$$A^T P + PA = -Q$$

And the above Theorem becomes

If A is Hurwitz, then for any Pos Def Q, there exists Pos Def P such that

$$A^T P + PA = -Q$$

Side Note: if we view the Lyap Eq'n as an arbitrary Matrix DE:

$$\dot{P}(t) + A^T(t)P(t) + P(t)A(t) = -Q(t)$$

The corresponding sol'n is

$$P(t) = \int_t^\infty \Phi^T(\tau, t)\, Q(\tau)\, \Phi(\tau, t)\, d\tau$$

STM for $A(t)$

We again need to be careful with the wording!

The above does <u>not</u> imply that for any P, the corresponding Q is positive definite!

For example, $P = I$ ($V(x) = \|x\|_2^2$) might not yield a pos def Q even when A is stable.

However, there are infinitely many other choices $P \neq I$ for which the corresponding Q would be pos. def.

Solving for P given Q and A for LTI systems is a well studied problem in linear algebra.

Matlab's "lyap" function can solve for P given Q and A.

Now what if a particular P, Q give inconclusive results? i.e. you can find $V(x,t) = x^T P(t) x$ for which

$$\dot{V} = -\alpha(t)\|x(t)\|_2^2$$

But $\alpha(t)$ is not ≥ 0 generally. (maybe even $\alpha(t) < 0$!)

Here you can:

1.) Try different $P(t), Q(t)$

2.) Try to investigate further:

i.e.

$$\dot{V} \leq -\alpha(t)\|\underline{x}(t)\|_2^2$$

$$\Rightarrow \dot{V} \leq \left(\frac{-\alpha(t)}{\rho}\right) V = \dot{\omega} = -\frac{\alpha(t)}{\rho}\omega$$

$$\Rightarrow \omega(t) = \omega(t_0)\, e^{-\frac{1}{\rho}\int_{t_0}^{t} \alpha(\tau)\,d\tau}$$

(as in the proof of UES, where again $\rho = \max_t \lambda_{max}(P(t))$.)

Then from the comparison lemma

$$V(\underline{x}(t), t) \leq V(\underline{x}_0, t_0)\, e^{-\frac{1}{\rho}\int_{t_0}^{t} \alpha(\tau)\,d\tau}$$

and we can investigate the behavior of the integral to see if we can ensure boundness or convergence.

"Local" Stability of Linear Systems

Recall that the homog. dynamics actually satisfy

$$\dot{\underline{x}}(t) = A(t)\underline{x}(t) + \underline{r}(\underline{x}(t), t)$$

where $\underline{r}(\underline{x}(t), t)$ is the truncation of the Taylor series expansion of the original nonlinear state equation function $\underline{f}$.

Can the presence of this term Radically change our conclusions of stability BASED on $A(t)$?

Generally no, As long as we can ensure $\|\underline{x}(t)\|$ remains sufficiently small, but then this means that our stability conclusions are necessarily limited to such (sufficiently small) $\underline{x}$. Thus these conclusions are *local* (holding only for a restricted set of $\underline{x}$) and not *global* (holding for arbitrary $\underline{x}$).

But how small must this local set be? In particular what is the "region of convergence" within which our stability analysis will be unaffected by $\underline{r}(\underline{x}, t)$?

Lyapunov gives us the tools to quantify this region.

Including the remainder term, we now have $V = x^T P x$

has derivative

$$\dot{V} = -x^T Q x + 2 x^T P r$$

where again

$$-Q = \dot{P} + A^T P + P A.$$

If $Q(t)$ is unif. PD (so $A(t)$ is UES) then as before

$$\dot{V} \leq -\nu \|x\|_2^2 + 2 x^T P r$$

and we need to quantify the impact of the added term

We can use a worst case Bound

$$x^T P r \leq |x^T P r|$$

$$\leq \|x\| \|P\|_i \|r\|$$

Let $\gamma_p = \max_t \|P(t)\|_i < \infty$ (since $P(t)$ is Bounded)

Then

$$\dot{V} \leq -\nu \|x\|_2^2 + 2\gamma_p \|x\| \|r\|$$

So we have a (negative) quadratic term in $\|x\|$, together with a (positive) Linear term in $\|x\|$

When does the negative "overpower" the positive?